As we've seen, the general uncertainty principle puts a quantitative limit on the simultaneous uncertainties of two observables **A** and **B**. This idea was captured in Inequality 5.13:

$$\Delta \mathbf{A}\ \Delta \mathbf{B} \geq \frac{1}{2}|\langle\Psi|[\mathbf{A}, \mathbf{B}]|\Psi\rangle|.$$

Now let's apply this principle directly to the position and momentum operators **X** and **P**. In this case, the commutator is just a number and its expectation is that same number. Replacing **A** and **B** with **X** and **P** gives

$$\Delta \mathbf{X}\ \Delta \mathbf{P} \geq \frac{1}{2}|\langle\Psi|[\mathbf{X}, \mathbf{P}]|\Psi\rangle|,$$

and replacing [**X, P**] with $i\hbar$ results in

$$\Delta \mathbf{X}\ \Delta \mathbf{P} \geq \frac{1}{2}|i\hbar\langle\Psi|\Psi\rangle|.$$

But <Ψ|Ψ> equals 1, and the end result is

$$\Delta \mathbf{X}\ \Delta \mathbf{P} \geq \frac{1}{2}\hbar.$$

No experiment can ever beat this limitation. You can try your best to determine a particle's momentum and position simultaneously in a reproducible manner, but no matter how careful you are, the uncertainty in the position times the uncertainty in the momentum will never be less than ½$\hbar$.

As we saw in Section 8.2.1, the wave function of an eigenstate of **X** is highly concentrated about some point x_0; in this eigenstate, the probability is also perfectly localized. On the other hand, the probability $P(x)$ for a